

ZEWAIL CITY OF SCIENCE AND
TECHNOLOGY

Faculty of Engineering

RAPID THERMAL PROCESSING (RTP)

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THERMODYNAMICS

NANENG 218

OUR CONTENT



INTRODUCTION

Rapid Thermal Processing (RTP) is an essential technique in modern semiconductor manufacturing. Unlike traditional furnace annealing, which heats wafers for long periods, RTP uses strong infrared (IR) radiation to heat a semiconductor wafer very quickly — usually within a few seconds — to temperatures above 1000°C , followed by rapid cooling. This precise heating process helps reduce unwanted dopant diffusion, lowers the overall thermal budget, and improves device performance and manufacturing yield.

This report explores the role of RTP in semiconductor fabrication, including its operating principles, system design methods, major engineering challenges, and the main components of an RTP system. In addition, a simplified 2D thermal model of an RTP chamber is simulated using COMSOL Multiphysics to study the infrared heating process and the steady-state temperature distribution across a silicon wafer.

2. RESEARCH ON RTP IMPORTANCE, CONCEPTS, DESIGN APPROACHES, AND KEY CHALLENGES

2.1 Importance of RTP in Semiconductor Manufacturing

As semiconductor device dimensions continue to shrink to sub-5 nm technology nodes, the precision requirements for thermal processing become increasingly demanding. Rapid Thermal Processing (RTP) addresses several critical needs in modern VLSI (Very Large-Scale Integration) fabrication:

- 1. Dopant Activation:** After ion implantation, dopants must be electrically activated through annealing. RTP performs this process within seconds, minimising unwanted dopant diffusion and preserving the sharp junction profiles required for short-channel transistors.
 - 2. Gate Dielectric Growth:** Ultra-thin gate oxides such as SiO_2 and high-k dielectric materials require highly accurate temperature control during oxidation. RTP enables precise dielectric growth while reducing contamination and defects.
 - 3. Silicide Formation:** Metal silicides such as TiSi_2 , NiSi , and CoSi_2 are formed using rapid annealing in RTP systems. These silicides create low-resistance ohmic contacts while minimising damage to nearby device structures.
 - 4. Thermal Budget Reduction:** In semiconductor fabrication, the total thermal budget (temperature $\times$ processing time) must be minimized to avoid device degradation and unwanted diffusion effects. RTP significantly lowers the thermal budget compared to conventional furnace processing because of its extremely short heating cycles.
 - 5. Single-Wafer Processing:** RTP systems process wafers individually rather than in large batches. This allows better real-time process control, faster recipe adjustments, and improved production traceability in advanced semiconductor fabrication facilities.
- Overall,** RTP has become indispensable in advanced CMOS, memory technologies such as DRAM and Flash, and power semiconductor fabrication, where strict control of the wafer's thermal history is essential for achieving high performance and reliability.

2.2 FUNDAMENTAL CONCEPTS OF RTP

2.2.1 Heating Mechanism

- Main Heat Transfer Mechanism: RTP mainly uses thermal radiation for wafer heating.
- IR Heating Source: Tungsten-halogen or arc lamps emit strong infrared radiation in the 0.5–5 μm wavelength range.
- Wafer Absorption: Silicon wafers absorb IR energy efficiently, especially above 600°C, due to increased free-carrier absorption.

- Heat Transfer Law: Radiative heating in RTP follows the Stefan–Boltzmann law.

At temperatures exceeding 1000°C, radiative heating becomes the dominant mode of heat transfer, accounting for nearly 90% of the total thermal energy transfer inside the RTP chamber. This dominance of radiation enables the extremely rapid heating and cooling rates required for advanced semiconductor processing.

$$q = \epsilon \sigma T^4$$

2.2.2 Rapid Heating and Cooling

- Rapid Heating: RTP heats silicon wafers very quickly at rates of 10–250°C/s using high-power infrared lamps and the wafer's low thermal mass.
- Rapid Cooling: After processing, wafers cool at 50–150°C/s mainly through radiation and convection.
- Process Advantage: Fast heating and cooling reduce unwanted dopant diffusion, prevent secondary reactions, and maintain the desired material microstructure.

2.2.3 Design Approaches

- Single-Zone Lamp Arrays: Simple RTP design with uniform lamp distribution for basic processing needs.
- Multi-Zone Lamp Arrays: Independently controlled lamp zones improve temperature uniformity across the wafer.
- Rotating Susceptor: Wafer rotation reduces heating non-uniformities and improves temperature accuracy.
- Reflective Chamber Design: Gold or polished aluminium walls reflect IR radiation for more uniform heating.
- Pyrometric Control: Optical pyrometers monitor wafer temperature in real time for closed-loop control.

2.3 KEY CHALLENGES IN RTP

2.3.1 Temperature Uniformity

WIWU is a key challenge in RTP due to edge cooling, which creates center-to-edge temperature gradients. This leads to non-uniform dopant activation, crystal defects, and device variation. It is reduced using multi-zone heating, wafer rotation, and edge/guard rings to improve temperature uniformity.

2.3.2 Wafer Slip and Thermal Stress

Rapid RTP ramping causes thermal stress that can exceed silicon's strength at high temperatures, leading to slip defects, wafer warpage, and device failure. This is reduced by controlling ramp rates and using multi-zone heating.

2.3.3 Emissivity-Dependent Measurement Errors

Pyrometric temperature measurement depends on accurate wafer emissivity, but silicon emissivity changes with temperature, doping, and surface conditions. This can cause temperature errors ($\pm 20\text{--}50^\circ\text{C}$), leading to process variations. To reduce this, dual-wavelength and emissivity-corrected pyrometers are used.

2.3.4 Process Scalability and Integration

As wafer size increases from 200 mm to 300 mm (and future 450 mm), maintaining temperature uniformity becomes more difficult due to the larger heating area. Also, integrating RTP into clustered multi-chamber systems requires careful design to ensure low contamination and fast wafer transfer, especially for in-situ processes like pre-cleaning and annealing.

2.4 RTP SYSTEM COMPONENTS AND MATERIAL CHOICES

Component	Material	Reason for Selection
IR Lamp Array	Tungsten-Halogen	Tungsten-halogen lamps emit strong broadband IR radiation (peaking around 1 μm) that is efficiently absorbed by silicon. The halogen cycle extends filament lifetime, while high power density (up to 100 W/cm²) enables rapid wafer heating in RTP systems.
Process Chamber	Stainless Steel / Aluminum	Stainless steel RTP chambers are strong, corrosion-resistant, and often polished or gold-coated to improve IR reflectivity and heating uniformity.
Quartz Window / Tube	Fused Quartz (SiO ₂)	Fused quartz is used in RTP because it transmits IR radiation efficiently (>90%), withstands high temperatures (up to ~1100°C), and offers good thermal shock resistance and chemical inertness.
Wafer Susceptor / Support	SiC-Coated Graphite	SiC-coated graphite combines a high-emissivity SiC surface (~0.9) for uniform heating with a graphite core that provides high thermal conductivity and low thermal mass. The SiC layer also prevents carbon contamination of the wafer.

2.4 RTP SYSTEM COMPONENTS AND MATERIAL CHOICES

Component	Material	Reason for Selection
Chamber Wall Coating	Gold or Polished Al	Gold coatings are used in RTP chambers because they provide very high IR reflectivity (>98%), which improves lamp energy efficiency and helps achieve better temperature uniformity across the wafer.
Temperature Sensor	Optical Pyrometer (Si or InGaAs)	Non-contact pyrometry measures wafer temperature by detecting its thermal radiation at specific wavelengths (e.g., 0.95 μm or 1.55 μm). It avoids contamination and enables real-time closed-loop temperature control.
Silicon Wafer	single-Crystal Silicon (p or n type)	The wafer substrate absorbs IR radiation efficiently. Its thin structure (~0.5 mm) gives low thermal mass, enabling fast ramp rates in RTP. Resistivity and crystal orientation are selected based on the specific process requirements.
Process Gas Lines	Electropolished 316L SS	Prevents contamination and corrosion under oxidizing or reducing atmospheres (O ₂ , N ₂ , H ₂ , Ar, etc.). Electropolishing minimizes surface roughness and particle generation

3. COMSOL MULTIPHYSICS SIMULATION FOR THERMAL ANALYSIS

3.1 Simulation Overview and Objectives

The simulation models a simplified 2D cross-section of an RTP chamber using COMSOL Multiphysics. The primary objectives are:

- To visualise the steady-state temperature distribution within the RTP chamber, including the silicon wafer, quartz tube, and surrounding chamber.
- To quantify the heating efficiency of the IR lamp array and its effect on wafer temperature.
- To analyse surface-to-surface radiation as the dominant heat transfer mechanism at elevated temperatures.
- To plot the temperature at the midpoint of the silicon wafer under stationary conditions.

3.2 MODEL GEOMETRY AND DIMENSIONS

The 2D cross-sectional model represents a circular RTP chamber with concentric structures. The geometry was built in COMSOL Multiphysics based on the required design dimensions.

Table 2: Model Geometry Dimensions	
Component	Dimension
RTP Chamber Radius	120 mm
IR Lamp Radius	10 mm
Quartz Tube Outer Radius	60 mm
Quartz Tube Inner Radius	55 mm
Silicon Wafer Width	50 mm
Silicon Wafer Thickness	0.5 mm

The IR lamps are arranged in a circular pattern at a radial distance of 80 mm from the chamber center, positioned between the quartz tube outer wall and the RTP chamber wall, as shown in Figure 1.

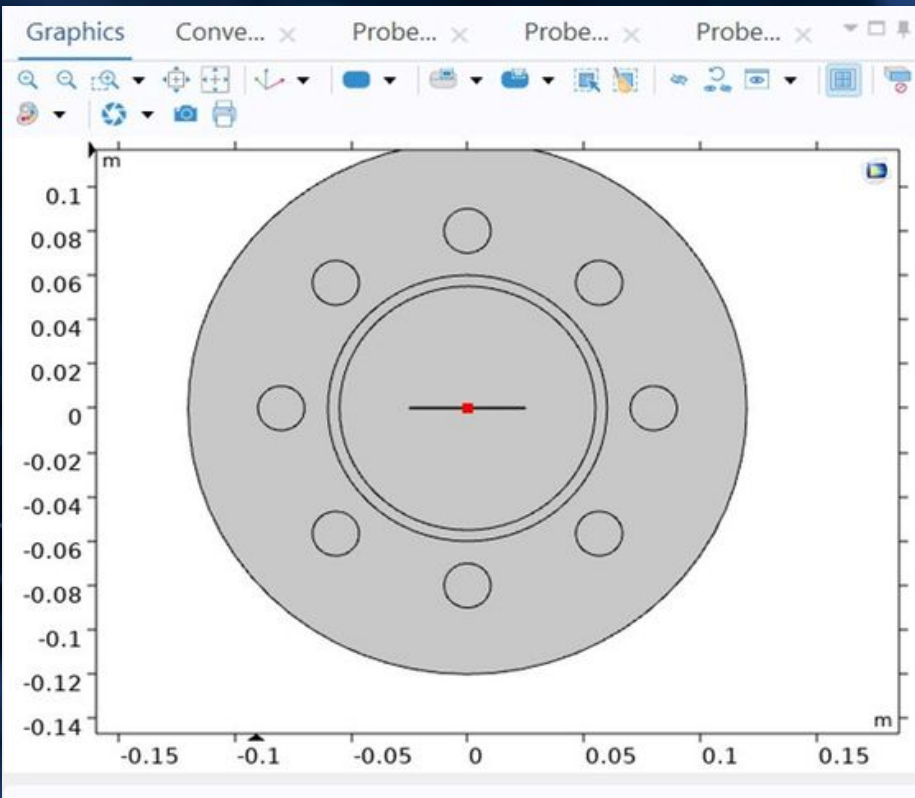


Figure 1: COMSOL model geometry — 2D cross-section of the RTP chamber showing the silicon wafer, quartz tube, and IR lamp arrangement.

3.3 BOUNDARY CONDITIONS AND PHYSICS SETUP

3.3.1 Heat Transfer Physics

The simulation uses the Heat Transfer in Solids module in COMSOL Multiphysics, coupled with Surface-to-Surface Radiation. Since RTP operates above 1000°C, radiation is the dominant heat transfer mode ($\approx 90\%$), governed by the Stefan–Boltzmann law, while conduction is also included within solid regions.

3.3.2 Boundary Conditions

The model applies the following boundary conditions in COMSOL Multiphysics:

- **Outer Chamber Wall ($R = 120\text{ mm}$):** Thermally insulated boundary ($q = 0$), assuming negligible heat loss due to strong insulation.
- **IR Lamps:** Each lamp is modelled with a 40 kW heat source and treated as a diffuse emitter with emissivity $\varepsilon = 0.99$.
- **Silicon Wafer:** Surface-to-surface radiation is applied with emissivity $\varepsilon = 0.5$, allowing radiative interaction with surrounding domains.
- **Quartz Tube:** Assigned built-in material properties; excluded from radiation boundaries since quartz primarily transmits and reflects IR rather than emitting it.
- **Interior Interfaces:** Standard heat flux continuity is enforced across all material boundaries.

3.3.3 Simulation Type

The simulation is performed as a steady-state (stationary) analysis, determining the equilibrium temperature field when the lamp heating is balanced by radiative heat losses inside the chamber.

3.4 MATERIAL PROPERTIES

Material properties for the simulation components are as follows:

Material	Thermal Cond. [W/(m·K)]	Density [kg/m³]	Heat Capacity [J/(kg·K)]	Emissivity
Silicon Wafer	Built-in (COMSOL)	Built-in (COMSOL)	Built-in (COMSOL)	0.5
IR Lamp	400	8700	10	0.99
Quartz Tube	Built-in (COMSOL)	Built-in (COMSOL)	Built-in (COMSOL)	Excluded

3.5 SIMULATION RESULTS AND ANALYSIS

3.5.1 Temperature Distribution

The COMSOL stationary simulation was successfully executed after meshing the geometry with a physics-controlled mesh (fine element size to resolve thermal gradients in the thin silicon wafer and near lamp surfaces). The following results were obtained:

The temperature distribution in the 2D model shows that the highest temperatures occur near the IR lamps, as expected. Heat reaches the silicon wafer mainly through radiation across the quartz/air gap, while the quartz tube allows most IR radiation to pass through with minimal absorption. The wafer attains a relatively uniform central temperature, with mild edge cooling effects observed at the periphery.

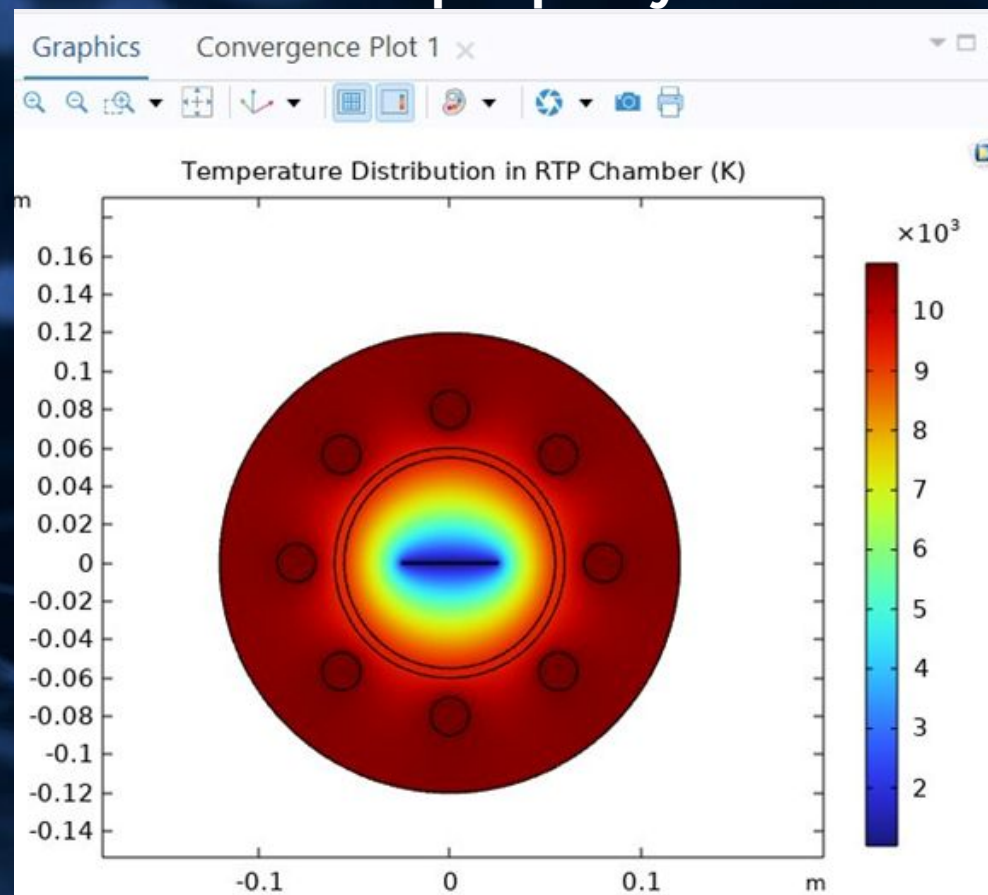


Figure 2: Temperature distribution in the RTP chamber (K) stationary simulation. Highest temperatures (red, $\sim 10,000$ K) are near the IR lamps; the silicon wafer reaches ~ 1027 K.

3.5 SIMULATION RESULTS AND ANALYSIS

3.5.2 Wafer Midpoint

The midpoint temperature of the silicon wafer was extracted from the simulation under steady-state conditions. With each lamp supplying 40,000 W, the wafer reaches a stable temperature governed by the balance between radiative heating from the lamps and thermal losses. The resulting temperature profile confirms that the IR lamp setup is effective in achieving the high temperatures required for dopant activation and annealing processes.

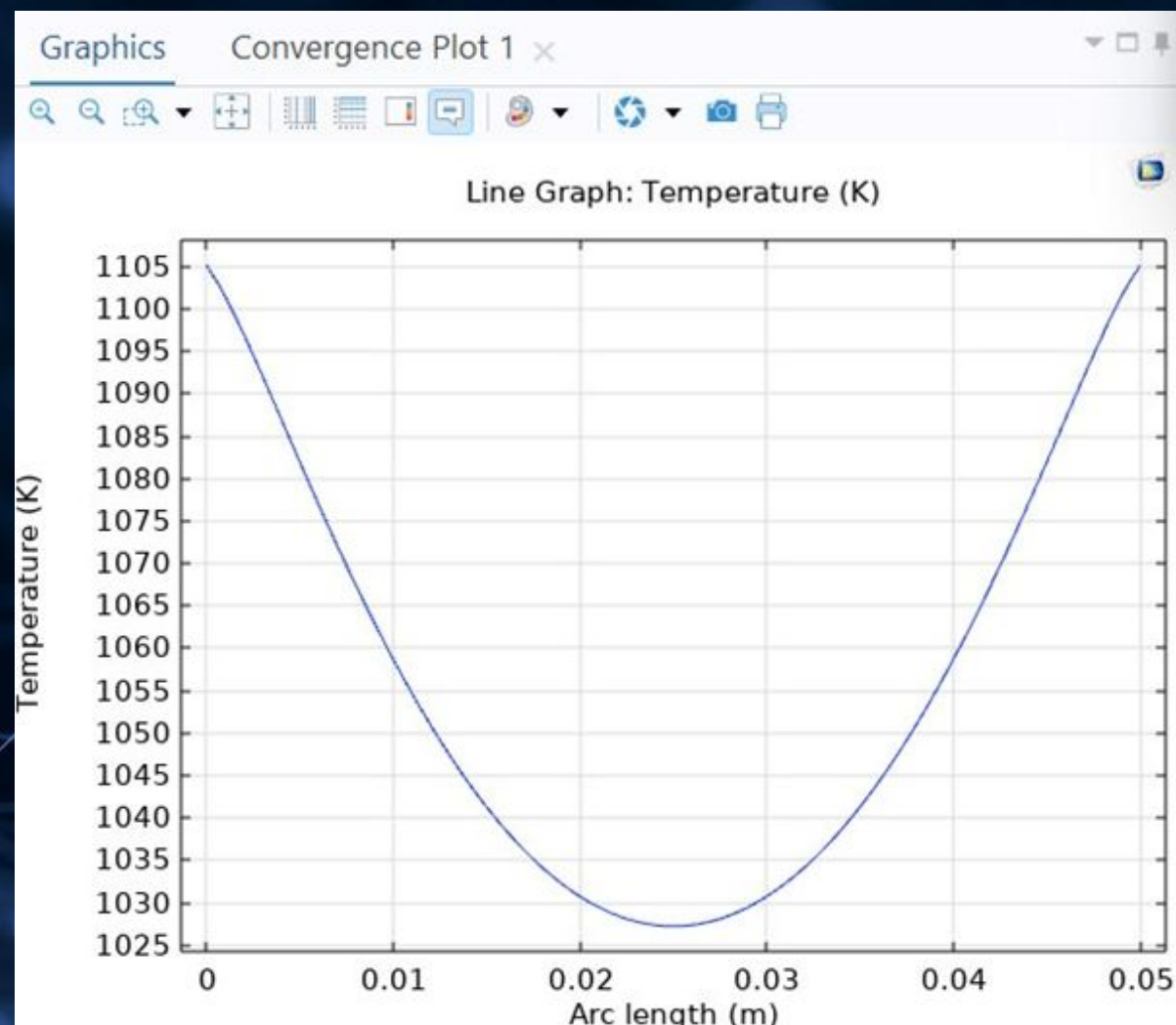


Figure 3: Temperature profile along the silicon wafer arc length, with minimum temperature at the wafer midpoint (~1027 K), rising toward the edges (~1105 K).

3.5 SIMULATION RESULTS AND ANALYSIS

3.5.3 Heating Efficiency

The simulation confirms that at temperatures above 1000°C, surface-to-surface radiation is the dominant heat transfer mechanism in the RTP chamber. The high lamp emissivity ($\epsilon = 0.99$) enhances radiative emission, while the wafer emissivity ($\epsilon = 0.5$) controls how much radiation is absorbed. Overall, the results validate the use of the Heat Transfer with Surface-to-Surface Radiation model in COMSOL Multiphysics for accurately capturing RTP thermal behavior.

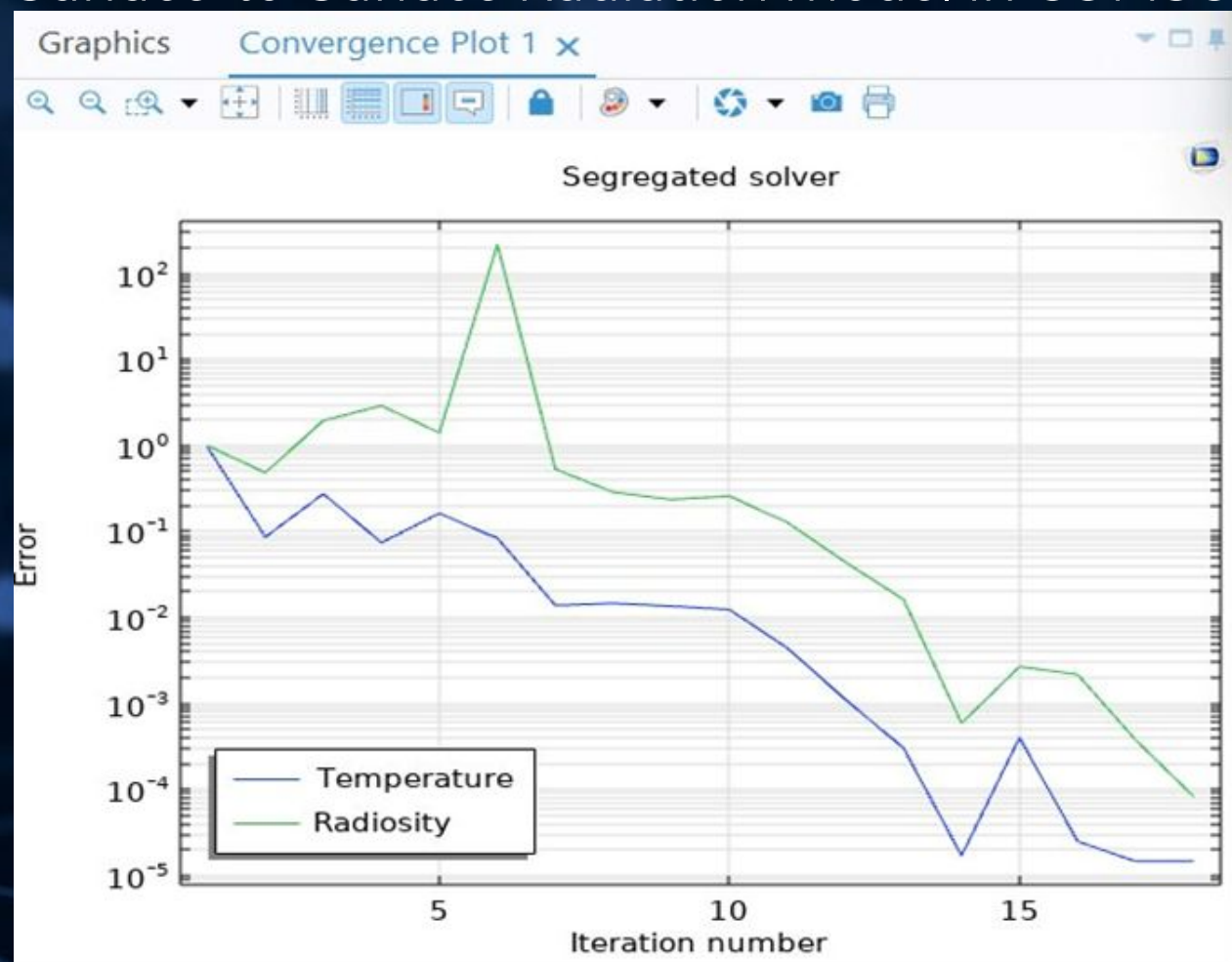


Figure 4: Segregated solver convergence plot showing Temperature and Radiosity residuals converging over 18 iterations

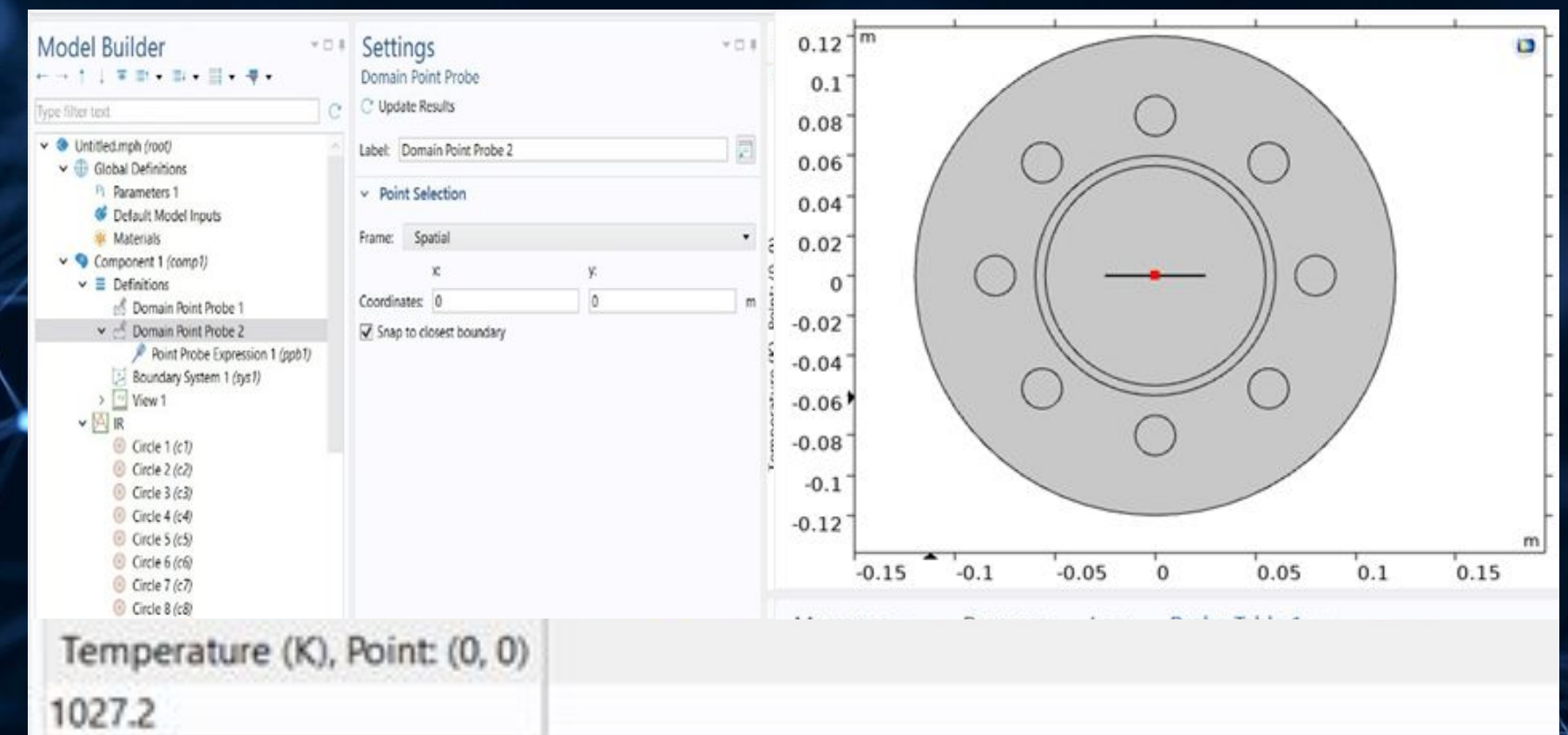


Figure 5: Domain Point Probe setup in COMSOL showing the probe at coordinate (0, 0) with a measured wafer temperature of 1027.2 K.

3.5 SIMULATION RESULTS AND ANALYSIS

3.5.4 temperature with time

The wafer temperature rises rapidly from room temperature due to the applied heating energy, showing a high heating rate in the first few seconds. After about 8–10 seconds, the temperature gradually stabilises near 1450 K as thermal equilibrium is reached, where heat input equals heat loss. The smooth and stable temperature profile indicates controlled heating, which is essential for uniform processing and reducing thermal stress in semiconductor fabrication.

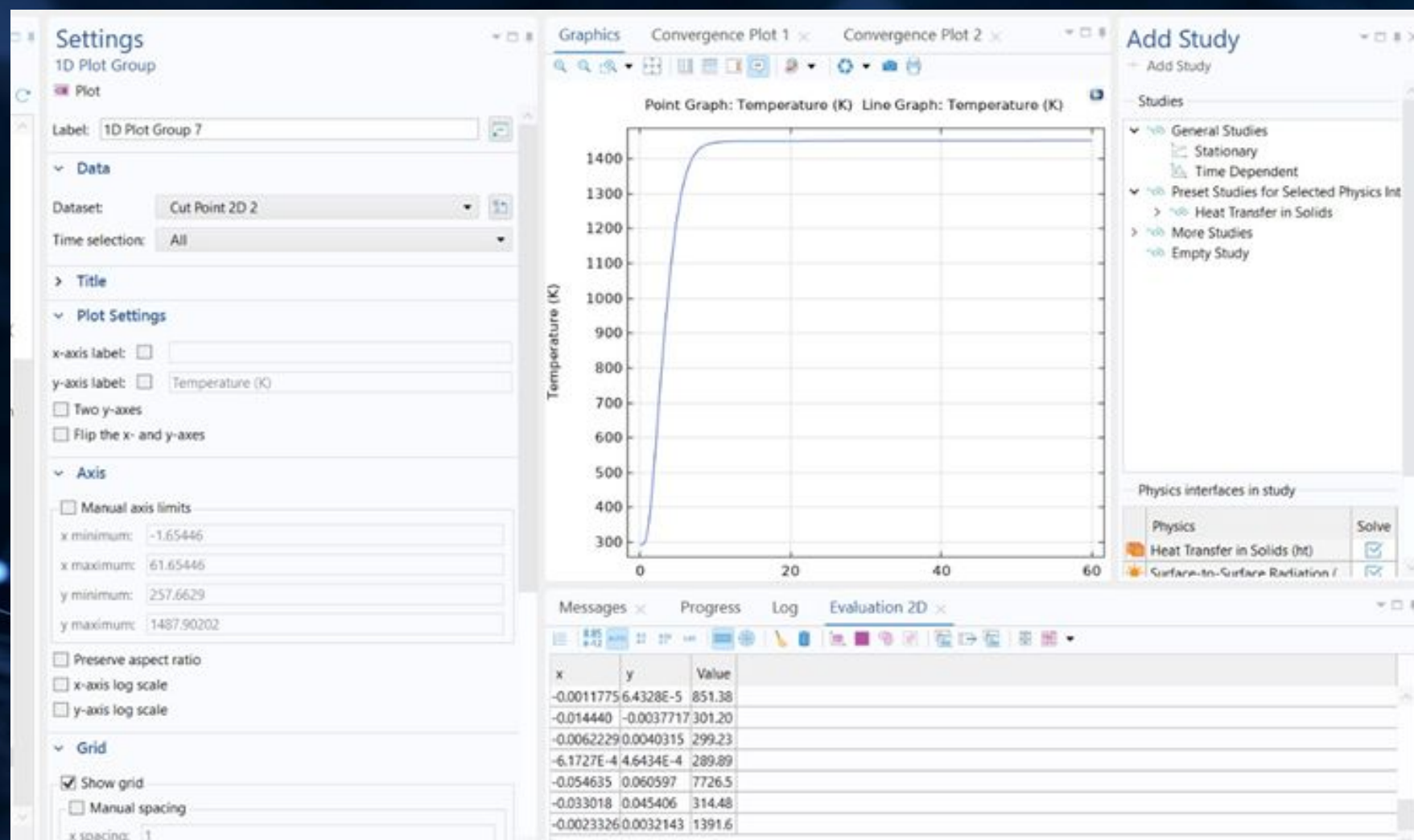


Fig. 6 shows the transient temperature response of the wafer during heating. The graph represents the wafer temperature at a selected point as a function of time..

4. CONCLUSION

This report presents a comprehensive study of Rapid Thermal Processing (RTP) and its role in modern semiconductor manufacturing. It covers the fundamental heat transfer mechanisms, system design approaches, material selection, and key operational challenges.

RTP is now a critical step in VLSI fabrication, enabling precise processes such as dopant activation, gate oxide formation, and silicide formation while minimising thermal budget and improving process control.

However, challenges such as temperature uniformity, pyrometric accuracy, wafer slip, and scalability continue to drive ongoing improvements in system design and control strategies.

A simplified 2D thermal model was developed using COMSOL Multiphysics to analyse RTP behaviour. The simulation confirms that radiative heat transfer dominates at high temperatures, consistent with the Stefan–Boltzmann law, and accurately captures IR lamp heating, surface-to-surface radiation, and steady-state wafer temperature distribution.

Overall, the results demonstrate that RTP simulation is a powerful tool for understanding thermal behaviour and optimising process design in advanced semiconductor manufacturing.

5. REFERENCES

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THANKS
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